UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Unit 1 · Partial Differentiation · all sessions (Class 001 – Class 011) · compiled 15 October 2026

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1 Short notes

1 · Functions of two or more variables

Quantities depending on several variables need functions of several variables (area $A(l,b)$, volume $v(r,h)$, potential $V(x,y,z,t)$, weather, pandemics). $z=f(x,y)$ assigns to each $(x,y)\in D$ a unique real number; D is the domain, the set of values the range. The graph is a surface.

2 · Limit of a function of two variables

$\lim_{(x,y)\to(a,b)} f=L$ if every path to (a,b) gives L. Polynomial: substitute. $\frac{0}{0}$: try paths ($y=0$, $y=x$, $y=mx$, $y=x^2$); two different values ⇒ no limit. $\frac{2xy}{3x^2+y^2}$: 0 on $y=0$, $\frac{1}{2}$ on $y=x$.

3 · Continuity of a function of two variables

Continuous at (a,b): $f(a,b)$ defined, limit exists, $\lim f=f(a,b)$; no hole or break in the surface. Not continuous ≠ no limit: $\frac{3x^2y}{x^2+y^2}$ has limit 0 but $f(0,0)$ undefined.

4 · Partial derivatives: history, definition and notation

Partial derivative: derivative in one variable, others constant. $z_x=\lim_{h\to0}\frac{f(x+h,y)-f(x,y)}{h}$; notation $\frac{\partial z}{\partial x}=z_x=f_x$. Leibniz (product rule), Legendre 1786, Jacobi 1841. All differentiation rules hold.

5 · Second-order and mixed partial derivatives

z_{xx}, z_{yy}, $z_{xy}=\frac{\partial}{\partial y}(z_x)=\frac{\partial^2 z}{\partial y\partial x}$, z_{yx}. Clairaut: $z_{xy}=z_{yx}$ when the second partials are continuous. Subscripts left to right, ∂ right to left.

6 · Higher-order partial derivatives

Differentiating $f(x, y)$ once gives the first-order partials f_x, f_y; differentiating again gives four second-order partials f_{xx}, f_{yy}, f_{xy}, f_{yx}. In general n differentiations give 2^n partials of order n for two variables and m^n for m variables. In f_{xy}, differentiate first in x, then in y. When the mixed partials are continuous, $f_{xy} = f_{yx}$ (Clairaut).

7 · The one-variable picture: derivative as slope

For $y = f(x)$, dy/dx measures how y changes as x changes. At x_0 it equals the slope of the tangent line to the curve at $(x_0, f(x_0))$, and the tangent is $y - f(x_0) = f'(x_0)(x - x_0)$.

8 · A function of two variables is a surface

A bivariate function $z = f(x, y)$ assigns one height to each point of the xy-plane, so its graph is a surface in 3-D. Points (x, y) on a circle or parabola in the xy-plane give a curve lying on that surface.

9 · First-order partials as slopes of tangents

A surface has infinitely many tangents at a point $P(x_0, y_0, f(x_0, y_0))$. $\partial f/\partial x$ at (x_0, y_0) is the slope of the tangent parallel to the x-axis. It is found by cutting the surface with the plane $y = y_0$ and differentiating the curve $f(x, y_0)$ at x_0. Likewise $\partial f/\partial y$ is the slope of the tangent parallel to the y-axis, from the curve $f(x_0, y)$ in the plane $x = x_0$.

10 · Slope in any other direction: the gradient

Tangents in other directions are handled by resolving the direction into x- and y-components. The resultant of $f_x\,\mathbf{i}$ and $f_y\,\mathbf{j}$ is the gradient ∇f. The slope along a unit vector $\hat{u}$ is $\nabla f \cdot \hat{u}$, and the largest slope is $|\nabla f|$.

11 · Second-order partials: concavity and mixed partials

Second-order partials describe bending. If $f_{xx} > 0$ the surface is concave up in the x direction; if $f_{yy} > 0$ it is concave up in the y direction (negative means concave down). The mixed partial f_{xy} tells how the rate of change of f in the x direction changes as we move in the y direction.

12 · Problems on partial derivatives

To find partials at a point, differentiate fully, holding the other variable constant, and substitute only at the end. To verify $f_{xy} = f_{yx}$, compute each side separately and compare. Watch the rule used: x^y is a power in x but an exponential in y.

13 · Composite functions and the one-variable chain rule

A composite function is a function of a function, e.g. $y = e^{\sin x^2}$. Its derivative is found from the outside in and the factors are multiplied: $e^{\sin x^2} \cdot \cos x^2 \cdot 2x$. For several variables the same idea gives the total derivative and the chain rule.

14 · The total derivative rule

If $z = f(x, y)$ and $x = x(t)$, $y = y(t)$, then z is a function of t alone and its total derivative is $\dfrac{dz}{dt} = \dfrac{\partial z}{\partial x}\dfrac{dx}{dt} + \dfrac{\partial z}{\partial y}\dfrac{dy}{dt}$. A tree diagram with one branch per intermediate variable gives the terms. The answer is written in terms of t or evaluated at the given t.

15 · The total differential: approximation and sensitivity

The total differential $dz = f_x\,dx + f_y\,dy$ (or $\sum f_{x_i} dx_i$) gives the approximate change in z for small changes in the variables. It approximates nearby values, $f(x_0 + dx, y_0 + dy) \approx f + f_x\,dx + f_y\,dy$. It also measures sensitivity: the larger coefficient marks the variable that matters most. It is used in substitution and error estimates.

16 · Rates of change problems

For a rate problem, write the quantity as a function of the changing variables, take d/dt of it with the total derivative rule, and substitute the given values and rates (decreasing means negative). If something stays constant its derivative is 0; a constraint between variables gives a missing rate by implicit differentiation.

17 · The chain rule for composite functions

If $z = f(x, y)$ with $x = g(u, v)$, $y = h(u, v)$, then z is a composite function of u, v and $z_u = z_x x_u + z_y y_u$, $z_v = z_x x_v + z_y y_v$ (one term per path in the tree diagram). With three intermediate variables there are three terms. If f is unknown, keep f_x, f_y as symbols. Write the final answer in the new variables. Standard results: $f(x - y, y - z, z - x)$ gives $u_x + u_y + u_z = 0$; in polar form $u_x^2 + u_y^2 = u_r^2 + u_\theta^2/r^2$.

18 · Implicit functions of two variables

An implicit function is given by a relation $f(x, y) = c$, a level curve. Along it $df = f_x\,dx + f_y\,dy = 0$, so $\dfrac{dy}{dx} = -\dfrac{f_x}{f_y}$ ($f_y \neq 0$). Direct differentiation of both sides gives the same result. The slope at (x_0, y_0) gives the tangent $y - y_0 = m(x - x_0)$; e.g. $xe^y + ye^x = 0$ has tangent $x + y = 0$ at the origin.

19 · Implicit functions of three variables

For $f(x,y,z)=0$, z is a function of x and y: $\dfrac{\partial z}{\partial x}=-\dfrac{f_x}{f_z}$ (hold y), $\dfrac{\partial z}{\partial y}=-\dfrac{f_y}{f_z}$, and $\dfrac{\partial y}{\partial x}=-\dfrac{f_x}{f_y}$ (hold z), provided the denominators are non-zero. Check that the point lies on the surface. For second derivatives, remember that z still depends on x, y.

20 · Homogeneous functions

$u=f(x,y)$ is homogeneous of degree n if $u=x^n g(y/x)$ or $y^n g(x/y)$; in three variables $x^n g(y/x, z/x)$. Equivalently $u(tx,ty)=t^n u$. A polynomial is homogeneous if all its terms have the same degree. The degree may be positive, negative, fractional or zero, e.g. $(\sqrt{x}+\sqrt{y})/(x+y)$ has degree $-\frac{1}{2}$.

21 · Euler's theorem

Euler's theorem: if $u=f(x,y)$ is homogeneous of degree n then $xu_x+yu_y=nu$; for $f(x,y,z)$, $xu_x+yu_y+zu_z=nu$. So only homogeneity and the degree are needed, not the derivatives. Proof: differentiate $u(tx,ty)=t^n u$ with respect to t and put $t=1$.

22 · Second and higher order: the corollary

Corollary: $x^2u_{xx}+2xyu_{xy}+y^2u_{yy}=n(n-1)u$; adding the first-order result gives n^2u. For a sum of homogeneous parts of different degrees apply this to each part, e.g. degrees 3 and 4 give $9u_1+16u_2$. Third order: $n(n-1)(n-2)u$.

23 · Extension: when f(u) is homogeneous

If $f(u)$ (not u) is homogeneous of degree n: $xu_x+yu_y=g(u)=nf(u)/f'(u)$ and $x^2u_{xx}+2xyu_{xy}+y^2u_{yy}=g(u)[g'(u)-1]$. For $u=\tan^{-1}\frac{x^4+y^4}{xy}$: $f=\tan$, $n=2$, $g=\sin 2u$, second order $\sin 2u(1-4\sin^2 u)$.

24 · Applications of homogeneous functions

Constants are homogeneous of degree 0 and linear functions of degree 1. Euler's theorem gives the scaling of a function; it is used in economics (production functions, marginal productivity, elasticity) and in thermodynamics (extensive variables degree 1, intensive degree 0). A function of ratios has degree 0, so its Euler sum is 0.

25 · Testing homogeneity: the scaling test

Scaling test: $u(\lambda x,\lambda y)=\lambda^n u(x,y)$ means degree n (positive, negative or zero). Slide examples: $\frac{x^4+2x^2y^2+y^4}{(x^2+y^2)^2}$ degree 0; $\frac{\sin^{-1}(y/x)}{\sqrt{x^2+y^2}}$ degree -1; $\frac{x^2y+xy^2}{(x+y)^3}$ degree 0; $\frac{e^x-e^y}{x-y}$ not homogeneous.

26 · Verifying Euler's theorem directly

To verify: find n, differentiate, form $xu_x + yu_y$, simplify to nu. For $Z = \frac{x+y}{\sqrt{x}+\sqrt{y}}$ ($n = \frac{1}{2}$): $xZ_x + yZ_y = \frac{(x+y)(\sqrt{x}+\sqrt{y})}{2(\sqrt{x}+\sqrt{y})^2} = \frac{1}{2}Z$.

27 · When u itself is homogeneous: first and second order

u homogeneous of degree n: $xu_x + yu_y = nu$, $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$, sum n^2u. $\frac{x-y}{x^2+y^2}$: $-u$, $2u$. ESA: $5u$ & $20u$; $9u$; $17\pi/2$; ratios only give 0.

28 · When f(u) is homogeneous: the extension

If $f(u)$ has degree n: $g = nf/f'$, second order $g(g'-1)$. $e^{x^3y^3/(x^2+y^2)}$: $4u\log u$. $\log\frac{x^2+y^2}{x+y}$: 1, -1. $\sin^{-1}\frac{x+y}{\sqrt{x}+\sqrt{y}}$: $\frac{1}{2}\tan u$, $\frac{1}{2}\tan u(\frac{1}{2}\sec^2 u - 1)$. $\sin^{-1}\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}$: $-3\tan u$. ESA: $\tan^{-1}\frac{x^4+y^4}{xy}$ gives $\sin 2u$, $\sin 2u(1-4\sin^2 u)$.

29 · Why Taylor's and Maclaurin's series?

Series turn complicated functions into polynomials, which can be evaluated, integrated and used in limits: $\int_0^1 \sin(x^2)dx \approx \frac{1}{3} - \frac{1}{42} + \frac{1}{1320} = 0.3103$; $\lim \frac{\sin x - x}{x^3} = -\frac{1}{6}$. Applications: integrals, asymptotics, differential equations, Lorentz factor, $\sin\theta \approx \theta$, optimization, visualization.

30 · Taylor's and Maclaurin's series in one variable

Taylor about a: $f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \ldots$; Maclaurin about 0. Needs finite derivatives at the point. Standard: e^x, $\log(1+x)$, $(1+x)^n$, $\sin$, $\cos$, $\sinh$, $\cosh$, $\tan^{-1}$, $\sin^{-1}$, $\cos^{-1}$; $\log(1+x)$ and $(1+x)^n$ only for $|x| < 1$.

31 · Taylor's expansion of a function of two variables

About (a,b) with $h = x-a$, $k = y-b$: $f + \{hf_x + kf_y\} + \frac{1}{2!}\{h^2f_{xx} + 2hkf_{xy} + k^2f_{yy}\} + \frac{1}{3!}\{h^3f_{xxx} + 3h^2kf_{xxy} + 3hk^2f_{xyy} + k^3f_{yyy}\}$, all at (a,b). ESA: $\sin x \sin y$ about $(\frac{\pi}{4}, \frac{\pi}{4})$ gives $\frac{1}{2} + \frac{1}{2}(h+k) - \frac{1}{4}(h-k)^2 - \frac{1}{12}(h+k)^3$.

32 · Maclaurin's expansion of a function of two variables

About $(0,0)$: $f + \{xf_x + yf_y\} + \frac{1}{2!}\{x^2f_{xx} + 2xyf_{xy} + y^2f_{yy}\} + \ldots$ ESA: $\log(1+x-y) = (x-y) - \frac{(x-y)^2}{2} + \frac{(x-y)^3}{3}$; $e^{ax}\sin by = by + abxy + \frac{a^2b}{2}x^2y - \frac{b^3}{6}y^3$; $\sin(x+2y) = (x+2y) - \frac{(x+2y)^3}{6}$.

33 · Taylor expansion about a point (a, b)

About (a,b): $h = x-a$, $k = y-b$; table of $f, f_x, f_y, f_{xx}, f_{xy}, f_{yy}$ at the point; $f + \{hf_x + kf_y\} + \frac{1}{2!}\{h^2f_{xx} + 2hkf_{xy} + k^2f_{yy}\}$. $e^x\cos y$ at $(1, \frac{\pi}{4})$: $\frac{e}{\sqrt{2}}[1 + h - k + \frac{1}{2}(h^2 - 2hk - k^2)]$. $\sin(xy)$ at $(1, \frac{\pi}{2})$: $1 - \frac{\pi^2}{8}h^2 - \frac{\pi}{2}hk - \frac{1}{2}k^2$. $xy^2 + y\cos(x-y)$ at $(1,1)$: $2 + h + 3k + \frac{1}{2}(-h^2 + 6hk + k^2)$.

34 · Maclaurin expansion about the origin

About $(0,0)$: $f + \{xf_x + yf_y\} + \frac{1}{2!}\{\dots\} + \frac{1}{3!}\{\dots\}$. $\log(2x+y+1) = (2x+y) - \frac{(2x+y)^2}{2}$; $e^{x-y} = 1 + (x-y) + \frac{(x-y)^2}{2} + \frac{(x-y)^3}{6}$; $e^x \log(1+y) = y + xy - \frac{y^2}{2} + \frac{x^2y}{2} - \frac{xy^2}{2} + \frac{y^3}{3}$. Check one-combination functions with the one-variable series.

35 · Approximate values from a Taylor polynomial

Expand about a nearby easy point and substitute small h, k with signs. $\tan^{-1}(xy)$ at $(0.9, -1.2)$ from $(1,-1)$: -0.835 (first order), -0.823 (second); exact -0.824. $\tan^{-1}\frac{0.9}{1.1} \approx 0.6854$.

36 · Maxima, minima and saddle points

Local maxima are peaks, local minima are pits; at both the tangent plane is horizontal. A saddle point also has a horizontal tangent plane but rises one way and falls another. A function can have several of each.

37 · Definitions: relative maximum, minimum and extreme value

Relative maximum at (a,b): $f(a,b) > f(a+h, b+k)$ for small h, k; minimum: $<$. Equivalently $\Delta = f(a+h,b+k) - f(a,b)$ has one sign: negative ⇛ maximum, positive ⇛ minimum. Either is an extreme value.

38 · Conditions and the working rule

Solve $f_x = 0$, $f_y = 0$. At each point find $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$. $rt - s^2 > 0$: $r < 0$ max, $r > 0$ min; $rt - s^2 < 0$: saddle; $rt - s^2 = 0$: doubtful (use Δ). Substitute the point in f for the value.

39 · The method, and the ESA cubic functions

$f_x = 0$, $f_y = 0$ → split into cases → stationary points → r, s, t → $rt - s^2$ and r → values. ESA: $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$: max 112 at $(4,0)$, min 108 at $(6,0)$, saddles $(5, \pm 1)$. $x^3 - 12x + y^3 + 3y^2 - 9y$: max 43 at $(-2,-3)$, min -21 at $(2,1)$, saddles $(2,-3)$, $(-2,1)$.

40 · Functions with many stationary points

$x^4 + y^4 - x^2 - y^2 + 1$: max 1 at $(0,0)$, min $\frac{1}{2}$ at $(\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}})$, four saddles on the axes. $\sin x + \sin y + \sin(x+y)$: $x = y = \frac{\pi}{3}$, max $\frac{3\sqrt{3}}{2}$.

41 · Applied problems: eliminate one variable first

Eliminate with the condition, then use the two-variable test. $400xyz^2$ on the unit sphere: max 50 at $(\pm\frac{1}{2}, \pm\frac{1}{2})$ (same signs); $kxyz^2$ on radius a: $\frac{ka^4}{8}$. Open box, volume V: $A = xy + \frac{2V}{x} + \frac{2V}{y}$, $x = y = \sqrt[3]{2V}$, $z = \frac{x}{2}$.

42 · Lagrange's method: the auxiliary function and the procedure

Optimise f subject to $\phi = 0$: $F = f + \lambda\phi$; solve $F_x = F_y = F_z = 0$ with $\phi = 0$ for x, y, z, λ. Advantages: no elimination; extends to many variables and constraints. Disadvantages: nature not decided; the equations are only necessary.

43 · Geometric problems: inscribed solids, nearest and farthest points

Box in sphere: $f = 8xyz \Rightarrow x = y = z = \frac{r}{\sqrt{3}}$, cube. Ellipsoid: $V_{\max} = \frac{8abc}{3\sqrt{3}}$. Distances: optimise the square. $x^2 - z^2 = 1$: nearest $(\pm 1, 0, 0)$. $5x^2 + 6xy + 5y^2 = 8$: distances 1 (min) and 2 (max). $(3, 4, 12)$ to unit sphere: 12 and 14.

44 · Applied problems: boxes, splitting a number, temperature

Open box, surface 432: $12 \times 12 \times 6$. $24 = 4 + 8 + 12$ for xy^2z^3. $400xyz^2$ on the unit sphere: 50. CW: $a^3x^2 + b^3y^2 + c^3z^2$ with $\sum \frac{1}{x} = 1$: $(a + b + c)^3$; $x^3 + 8y^3 + 64z^3$ with $xyz = 1$: 24.

2 Keyword index

185 terms; the number in brackets is the topic.

r, s, t (38). f_{xx}, f_{xy}, f_{yy}.

Acceleration (11). Second derivative of position with respect to time (d^2y/dt^2).

Advantage (42). No elimination; many variables and constraints.

Approximate value (31). Evaluate the truncated series at a nearby point.

Approximate value (35). Truncated series at a nearby point.

Approximation (15). Estimating f near a known point with $f + f_x\,dx + f_y\,dy$.

Arbitrary function (20). The g in $x^n g(y/x)$: any function of the ratios.

Auxiliary function (42). $F = f + \lambda\phi$.

Bivariate function (1). $z = f(x, y)$, two independent variables.

Bivariate function (8). A function of two independent variables, $z = f(x, y)$.

Branch diagram (18). Diamond diagram for dz/dx when $y = h(x)$: $f_x \cdot 1 + f_y\,y'$.

Case split (39). $6y(x-5) = 0$: $y = 0$ or $x = 5$.

Chain (of a problem) (17). What depends on what; decides the formula.

Chain rule (one variable) (13). Differentiate outside-in and multiply.

Chain rule (several variables) (17). $z_u = z_x x_u + z_y y_u$.

Chain rule step (28). $F_x = f'(u)u_x$; then divide by $f'(u)$.

Clairaut's theorem (5). $z_{xy} = z_{yx}$ if the second partials are continuous.

Clairaut's theorem (6). Continuous mixed partials are equal: $f_{xy} = f_{yx}$.

Common factor (33). $\frac{e}{\sqrt{2}}$ in $e^x \cos y$ at $(1, \frac{\pi}{4})$.

Composite function (13). Function of functions, e.g. $f(x(u,v), y(u,v))$.

Concave down (in x) (11). The trace along x bends downward; $f_{xx} < 0$.

Concave up (in x) (11). The trace along x bends upward; $f_{xx} > 0$.

Constant function (24). Homogeneous of degree 0.

Constrained extremum (42). Extremum of f on $\phi = 0$.

Constraint (16). Relation between variables used to find a missing rate.

Continued product (44). xy^2z^3 for first, square of second, cube of third.

Continuity (3). $\lim f = f(a, b)$; no hole or break.

Contour plot (11). Level curves of $z = f(x, y)$ drawn in the xy-plane; one view in the class demo.

Corollary (Euler) (22). $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$.

Curve on a surface (8). The points $(x, y, f(x, y))$ above a curve in the xy-plane.

Cylinder (18). $x^2 + y^2 = c$; its layers are level surfaces.

Decreasing rate (16). A negative derivative with respect to time.

Definite integral by series (29). Integrate the series term by term.

Degree (20). The n in $x^n g(y/x)$.

Degree (25). The power n of λ; positive, negative, fractional or zero.

Degree 0 (24). Depends on ratios only; Euler sum 0 (not necessarily constant).

Degree 0 (25). Depends on ratios only; Euler sum 0.

Derivative (7). Rate of change; geometrically, the slope of the tangent.

Differential (15). Another name for the total derivative dz.

Directional derivative (10). Slope along a unit vector $\hat{u}$: $\nabla f \cdot \hat{u}$.

Disadvantage (42). Nature not determined; conditions only necessary.

Domain (1). The set D of pairs (x, y).

Doubtful case (38). $rt - s^2 = 0$: test gives no information.

Economics (24). Production functions, marginal productivity, elasticity of demand.

Elimination (41). Use the constraint to remove one variable.

Equipotential surface (18). Level surface on which the potential is constant.

Error estimate (15). Use of the differential, covered at the end of the unit.

Euler's theorem (21). $xu_x + yu_y = nu$ for homogeneous u of degree n.

Euler's theorem (27). $xu_x + yu_y = nu$.

Expansion point choice (35). Near the target, with exact values.

Explicit function (18). $y = g(x)$ given by a formula.

Extension (28). $xu_x + yu_y = nf(u)/f'(u)$ when $f(u)$ is homogeneous.

Extension of Euler's theorem (23). $xu_x + yu_y = nf(u)/f'(u)$ when $f(u)$ is homogeneous.

Extensive variable (24). Scales with the system: degree 1 (mass, volume).

Extreme value (37). A maximum or a minimum value.

Extreme values (39). f at the maxima and minima.

First principles (4). $z_x = \lim_{h\to 0} \frac{f(x+h,y)-f(x,y)}{h}$.

First-order partial (6). f_x or f_y: obtained by differentiating once.

Four equations (42). $F_x = F_y = F_z = 0$, $\phi = 0$.

g(g′−1) (28). Second-order sum for the extension.

g(u) (28). $nf(u)/f'(u)$.

Gradient (10). $\nabla f = f_x\,\mathbf{i} + f_y\,\mathbf{j}$; resultant of the axis slopes.

Hessian (38). $\begin{vmatrix} r & s \\ s & t \end{vmatrix} = rt - s^2$.

Higher-order partial (6). A partial obtained by differentiating 3, 4, ... n times.

Highest temperature (41). $400xyz^2$ on the unit sphere: 50.

Homogeneous differential equation (20). Solved with $y = vx$; uses the same idea.

Homogeneous function (20). $u(tx, ty) = t^n u$.

Homogeneous polynomial (20). All terms of the same total degree.

Horizontal tangent plane (36). Geometric meaning of a stationary point.

Hyperbolic cylinder (43). $x^2 - z^2 = 1$; nearest points $(\pm 1, 0, 0)$.

Hyperbolic paraboloid (16). Saddle surface $x^2/16 - y^2/9 = z$ (the slide says paraboloid).

Identity for any f (17). A result like $u_x + u_y + u_z = 0$ that holds whatever the outer function f is.

Implicit differentiation (18). Finding dy/dx from $f(x, y) = 0$: $-f_x/f_y$.

Implicit function (18). Function given by a relation $f(x, y) = c$.

In powers of (x − a) (31). About the point (a, b).

In powers of (x − a), (y − b) (33). About the point (a, b).

Increment signs (35). $h = 0.9 - 1 = -0.1$.

Independent variable (8). An input (x or y) that can be varied freely.

Indeterminate form (2). $\frac{0}{0}$ on substitution; try paths.

Inscribed box (43). $V = 8xyz$; cube in a sphere.

Integration by substitution (15). Uses the differential: $t = x^2$, $dt = 2x\,dx$.

Intensive variable (24). Unchanged by scaling: degree 0 (pressure, temperature).

Intermediate variable (14). Variable between z and the final variables.

Isotherm (18). Level surface of constant temperature.

Jerk (11). Third derivative of position with respect to time.

Lagrange multiplier λ (42). Undetermined constant multiplying the constraint.

Legendre (1786), Jacobi (1841) (4). Created and reintroduced the symbol ∂.

Leibniz (4). First used partial differentiation (product rule).

Level curve (18). $f(x,y)=c$; its slope is $-f_x/f_y$.

Level surface (19). $f(x,y,z)=c$, or $z=f(x,y)=c$.

Limit (2). Same value L along every path to (a,b).

Linear function (24). Homogeneous of degree 1 (no constant term).

Local (relative) maximum (37). $f(a,b)>f$ at all nearby points.

Local (relative) minimum (37). $f(a,b)<f$ at all nearby points.

Lorentz factor (29). Approximated by its Maclaurin series at low speed.

Maclaurin's series (30). Taylor's series about the origin.

Maclaurin's series (34). About the origin.

Maclaurin's series in two variables (32). E32.1 about $(0,0)$.

Mixed partial (5). z_{xy}, z_{yx}.

Mixed partial (6). f_{xy} or f_{yx}: differentiate with respect to different variables.

Mixed partial (22). $u_{xy}=u_{yx}$, giving the $2xy$ term.

Mixed partials (31). $f_{xy}=f_{yx}$; give the coefficient 2.

Multivariable function (1). A quantity depending on two or more variables.

Multivariable function (6). A function of two or more independent variables.

n(n−1)u (22). Right side of the second-order corollary.

Neighbourhood (37). Points $(a+h,b+k)$ with h,k small.

Not homogeneous (25). No single λ^n factors out (e.g. e^x-e^y).

nth-order partial (6). Obtained by n differentiations; 2^n of them for two variables.

n²u (27). First plus second order.

One-combination check (34). $f=g(px+qy)$: use g's series.

One-combination shortcut (32). $f(x+2y)$: use the one-variable series.

Open box (41). $A=xy+2yz+2zx$; $x=y=\sqrt[3]{2V}$, $z=\frac{x}{2}$.

Open box (44). Surface $xy+2yz+2zx$.

Order of accuracy (35). More terms, smaller error.

Order of subscripts (5). Left to right = order of differentiation.

Parallel gradients (42). $\nabla f=-\lambda\nabla\phi$ at the optimum.

Parameter (14). The single variable t in the total derivative.

Partial derivative (4). Derivative in one variable, others constant.

Partial derivative (6). Derivative with respect to one variable, the others held constant.

Partial direction (9). The variable (x or y) along which the tangent is drawn.

Path (2). A curve (line, parabola, ellipse) along which $(x,y)\to(a,b)$.

Pendulum approximation (29). $\sin\theta\approx\theta$ (one term).

Plane section (trace) (9). Curve cut from the surface by $y=y_0$ or $x=x_0$.

Point of tangency (9). $P=(x_0,y_0,f(x_0,y_0))$, where the tangents touch the surface.

Polar coordinates (17). $x=r\cos\theta$, $y=r\sin\theta$.

Polynomial approximation (29). Truncating the series after a few terms.

Product of series (34). $e^x\log(1+y)$, $e^{ax}\sin by$.

Quotient degree (25). Numerator degree minus denominator degree.

Quotient rule (26). Used to differentiate Z on slide 006·3.

Radius of validity (30). $|x|<1$ for $\log(1+x)$, $(1+x)^n$, $\frac{1}{1-x}$.

Range (1). The set of values f takes on.

Rate of change (7). How fast the output changes per unit change of an input.

Rate of change (16). Derivative with respect to time.

Ratio (20). y/x, z/x: the variables of g.

Related rates (16). Rates linked through a formula or a constraint.

Removable discontinuity (3). Limit exists, $f(a,b)$ undefined or different.

Resultant (10). Vector sum of components; gives the gradient from f_x, f_y.

Saddle (11). Surface concave up in one direction and down in another, e.g. $x^2 - y^2$.

Saddle point (36). Stationary, neither max nor min; $rt - s^2 < 0$.

Saddle point (39). $rt - s^2 < 0$; no extreme value.

Scaling (20). Multiplying all variables by t; multiplies u by t^n.

Scaling test (25). $u(\lambda x, \lambda y) = \lambda^n u$.

Second-order corollary (27). $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$.

Second-order partial (5). z_{xx}, z_{yy}.

Second-order partial (6). $f_{xx}, f_{yy}, f_{xy}, f_{yx}$: obtained by differentiating twice.

Sensitivity (15). Which variable's change affects the output most (largest coefficient in dz).

Separable function (40). f_x in x only, f_y in y only.

Single-valued function (7). Gives exactly one output for each input.

Slope (7). Rise per unit run of a line; for a tangent, the derivative.

Sphere (19). $x^2 + y^2 + z^2 = c$; nested level surfaces.

Squared distance (43). Optimise $x^2 + y^2 + z^2$ instead of the distance.

Standard series (30). e^x, $\log(1+x)$, $\sin x$, ... (slide 007·5).

Stationary / critical point (38). $f_x = f_y = 0$.

Stationary points (39). Solutions of $f_x = f_y = 0$.

Sum of homogeneous parts (22). Apply Euler to each part with its own degree.

Sum of homogeneous parts (27). Each part with its own degree.

Surface (1). Graph of $z = f(x,y)$.

Surface (8). Graph of $z = f(x,y)$ in three dimensions.

Tangent line (7). Line touching a curve at a point with the curve's slope there.

Tangent line (18). $y - y_0 = m(x - x_0)$ with $m = dy/dx$.

Tangent line (30). First two terms of Taylor's series.

Tangent parallel to the x-axis (9). Tangent in the plane $y = y_0$; slope $f_x(x_0, y_0)$.

Tangent parallel to the y-axis (9). Tangent in the plane $x = x_0$; slope $f_y(x_0, y_0)$.

Tangent plane (31). First-degree Taylor polynomial in two variables.

Taylor's series (30). Polynomial form of f about a point.

Taylor's series about (a, b) (33). E33.1.

Taylor's series in two variables (31). E31.1 about (a,b).

Taylor's theorem (29). Basis of L'Hôpital's rule and the mean value theorems.

Third-degree terms (31). $\frac{1}{3!}\{h^3 f_{xxx} + 3h^2k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}\}$.

Third-order Euler (22). $\sum$ with coefficients 1, 3, 3, 1 $= n(n-1)(n-2)u$.

Third-order partials (33). Needed for third-degree terms.

Total derivative (14). $dz/dt = z_x\,x' + z_y\,y'$.

Total differential (15). $dz = f_x\,dx + f_y\,dy$.

Tree diagram (14). Root z, branches labelled with derivatives; multiply along, add across.

Trigonometric stationary points (40). Subtract the conditions: $\cos x = \cos y$.

Two-path test (2). Two different path values ⇒ no limit.

Undo the outer function (28). $\sin^{-1} \to \sin$, $\log \to e^u$, $e \to \log$, $\tan^{-1} \to \tan$.

Unit vector (10). Vector of length 1; fixes a direction $\hat{u} = (a,b)$.

Up to n-th degree (33). Keep all terms of total degree $\le n$.

Up to three terms (34). Three non-zero terms.

Vanishing derivatives (33). $\sin(xy)$ at $(1, \frac{\pi}{2})$ has $f_x = f_y = 0$.

Velocity (11). First derivative of position with respect to time.

Verification (26). Compute $xu_x + yu_y$ directly and show it equals nu.

Vertical branch (18). $dx/dx = 1$ in the branch diagram.

Working rule (38). Stationary points $\to r, s, t \to$ sign of $rt - s^2$, then r.

3 Rapid recall

1	[Class 001] Define a function of two variables.	A rule assigning to each $(x, y) \in D$ a unique real number $f(x, y)$.
2	[Class 001] Graph of $z = f(x, y)$?	A surface in three dimensions.
3	[Class 001] When does $\lim_{(x,y)\to(a,b)} f$ exist?	When every path to (a, b) gives the same value.
4	[Class 001] How do you show a limit does not exist?	Find two paths giving different values.
5	[Class 001] $\lim_{(x,y)\to(2,1)}(x^2 - 2xy + 3y^2 - 4x + 3y - 6)$?	-8 (and -6 at $(2, -1)$).
6	[Class 001] $\frac{2xy}{3x^2+y^2}$ along $y = mx$?	$\frac{2m}{3+m^2}$: no limit at the origin.
7	[Class 001] Three conditions for continuity?	$f(a, b)$ defined, limit exists, limit $= f(a, b)$.
8	[Class 001] Is $\frac{3x^2y}{x^2+y^2}$ continuous at $(0, 0)$?	No ($f(0, 0)$ undefined), though the limit is 0.
9	[Class 001] Not continuous ⇒ no limit?	No.
10	[Class 001] Definition of $\frac{\partial z}{\partial x}$?	$\lim_{h\to 0} \frac{f(x+h,y)-f(x,y)}{h}$, y constant.
11	[Class 001] Who created and who reintroduced ∂?	Legendre (1786); Jacobi (1841).
12	[Class 001] Earliest use of partial differentiation?	Leibniz's product rule.
13	[Class 001] $\frac{\partial^2 z}{\partial y \partial x}$ in subscripts?	z_{xy} (x first).
14	[Class 001] Clairaut's theorem?	$f_{xy} = f_{yx}$ if the second partials are continuous.
15	[Class 001] $f = e^{x^2+y^2+z^2}$: f_{xx} at $(-1, 1, -1)$?	$6e^3$.
16	[Class 001] n for $v = r^n(3\cos^2\theta - 1)$?	$n = 2$ or -3.
17	[Class 001] $u = x^2 \tan^{-1}\frac{y}{x} - y^2 \tan^{-1}\frac{x}{y}$: u_{xy}?	$\frac{x^2-y^2}{x^2+y^2}$.
18	[Class 002] How many second-order partials does $f(x, y)$ have?	Four: $f_{xx}, f_{yy}, f_{xy}, f_{yx}$.
19	[Class 002] How many third-order partials does $f(x, y)$ have?	$2^3 = 8$.
20	[Class 002] How many partials of order n for m variables?	m^n.
21	[Class 002] What does $f'(x_0)$ mean geometrically?	Slope of the tangent to $y = f(x)$ at x_0.
22	[Class 002] What is the graph of $z = f(x, y)$?	A surface in 3-D.
23	[Class 002] What do points on a circle in the xy-plane give on the surface?	A curve lying on the surface.

24	[Class 002] How many tangents does a surface have at a point?	Infinitely many.
25	[Class 002] What is $f_x(x_0, y_0)$ geometrically?	Slope of the tangent at P parallel to the x-axis.
26	[Class 002] Which plane contains that tangent?	The plane $y = y_0$.
27	[Class 002] What is $f_y(x_0, y_0)$ geometrically?	Slope of the tangent at P parallel to the y-axis.
28	[Class 002] How do you compute $f_x(x_0, y_0)$ as an ordinary derivative?	$\frac{d}{dx} f(x, y_0)$ at $x = x_0$.
29	[Class 002] Name the vector $f_x\,\mathbf{i} + f_y\,\mathbf{j}$.	The gradient ∇f.
30	[Class 002] Slope along unit vector (a, b)?	$a f_x + b f_y$.
31	[Class 002] What does $f_{xx} > 0$ tell you?	Concave up in the x direction.
32	[Class 002] What does $f_{yy} < 0$ tell you?	Concave down in the y direction.
33	[Class 002] What does f_{xy} measure?	Change of the x-slope as you move in y.
34	[Class 002] When is $f_{xy} = f_{yx}$?	When both are continuous (Clairaut).
35	[Class 002] $\partial(x^y)/\partial x$ and $\partial(x^y)/\partial y$?	$y\,x^{y-1}$ and $x^y \log x$.
36	[Class 002] Physical names of y', y'', y''' in time?	Velocity, acceleration, jerk.
37	[Class 002] Is a tangent parallel to the x-axis horizontal?	No; its slope is f_x, in the plane $y = y_0$.
38	[Classes 003–004] $\dfrac{d}{dx} e^{\sin x^2}$?	$2x\, e^{\sin x^2} \cos x^2$.
39	[Classes 003–004] Total derivative of $z = f(x, y)$, $x(t)$, $y(t)$?	$dz/dt = z_x\, dx/dt + z_y\, dy/dt$.
40	[Classes 003–004] Ordinary or partial: z of x, y, both of t?	Ordinary dz/dt.
41	[Classes 003–004] Ordinary or partial: z of x, y, both of u, v?	Partial $\partial z/\partial u$, $\partial z/\partial v$.
42	[Classes 003–004] Chain rule for $\partial z/\partial u$?	$z_x x_u + z_y y_u$.
43	[Classes 003–004] Total differential of $z = f(x, y)$?	$dz = f_x\, dx + f_y\, dy$.
44	[Classes 003–004] Total differential of $u = x/y$?	$(y\, dx - x\, dy)/y^2$.
45	[Classes 003–004] Approximate $f(2.1, -3.03)$ given $f = 6$, $f_x = 1.3$, $f_y = -0.6$ at $(2, -3)$?	6.148.
46	[Classes 003–004] Tank $r = 2$, $h = 10$: more sensitive to r or h?	r: $dV = 40\pi\, dr + 4\pi\, dh$.
47	[Classes 003–004] “Area remains constant” means?	$dA/dt = 0$.

48	[Classes 003–004] A side decreasing at 2 cm/s: its rate?	-2 cm/s.
49	[Classes 003–004] dz/dt for $z = x^2y^3$, $x = t^2$, $y = 2t^3$?	$104t^{12}$ (the slide's $60t^{12}$ is wrong).
50	[Classes 003–004] Cone with $r = h$, $dV/dt = 9$, $h = 6$: dh/dt?	$1/(4\pi) \approx 0.0796$ cm/s.
51	[Classes 003–004] dy/dx for $f(x, y) = c$?	$-f_x/f_y$, if $f_y \neq 0$.
52	[Classes 003–004] dy/dx for $x^2 + xy + y^2 = 7$?	$-(2x + y)/(x + 2y)$.
53	[Classes 003–004] Tangent to $xe^y + ye^x = 0$ at the origin?	$x + y = 0$.
54	[Classes 003–004] $\partial z/\partial x$ from $f(x, y, z) = 0$?	$-f_x/f_z$ (hold y).
55	[Classes 003–004] $\partial y/\partial x$ from $f(x, y, z) = 0$?	$-f_x/f_y$ (hold z).
56	[Classes 003–004] $z^3 + xy - y^2z = 6$ at $(0, 1, 2)$: z_x?	$-1/11$.
57	[Classes 003–004] $u = f(x - y, y - z, z - x)$: $u_x + u_y + u_z$?	0.
58	[Classes 003–004] $u = f(x/y, y/z, z/x)$: $xu_x + yu_y + zu_z$?	0.
59	[Classes 003–004] $u = f(r)$, $r = \sqrt{x^2 + y^2 + z^2}$: u_x?	$f'(r)\, x/r$.
60	[Classes 003–004] $u = \tan^{-1}(y/x)$ in polar form: u_θ?	1.
61	[Classes 003–004] $u = e^{x^2+y^2}$ in polar form: u_θ?	0.
62	[Classes 003–004] $u_x^2 + u_y^2$ in polar form?	$u_r^2 + u_\theta^2/r^2$.
63	[Class 005] Define a homogeneous function of degree n.	$u = x^n g(y/x)$ or $y^n g(x/y)$.
64	[Class 005] Scaling form of the definition?	$u(tx, ty) = t^n u(x, y)$.
65	[Class 005] Degree of $3x + 4y$?	1.
66	[Class 005] Degree of $x^3 + y^3 + z^3 + 3xyz$?	3.
67	[Class 005] Degree of $(\sqrt{x} + \sqrt{y})/(x + y)$?	$-\frac{1}{2}$.
68	[Class 005] Degree of a constant?	0.
69	[Class 005] Is every degree-0 function constant?	No: y/x has degree 0.
70	[Class 005] Euler's theorem (two variables)?	$xu_x + yu_y = nu$.
71	[Class 005] Euler's theorem (three variables)?	$xu_x + yu_y + zu_z = nu$.

72	[Class 005] Second-order corollary?	$x^2u_{xx}+2xyu_{xy}+y^2u_{yy}=n(n-1)u$.
73	[Class 005] First plus second order?	n^2u.
74	[Class 005] Third order right side?	$n(n-1)(n-2)u$.
75	[Class 005] $u=x^3y^2\sin^{-1}(y/x)$: degree, Euler sums?	5; $5u$ and $20u$.
76	[Class 005] Extension: xu_x+yu_y when $f(u)$ is homogeneous?	$g(u)=nf(u)/f'(u)$.
77	[Class 005] Extension, second order?	$g(u)[g'(u)-1]$.
78	[Class 005] $u=\tan^{-1}\frac{x^4+y^4}{xy}$: xu_x+yu_y?	$\sin 2u$.
79	[Class 005] Same u: second order?	$\sin 2u(1-4\sin^2 u)$.
80	[Class 005] $x^3\sin^{-1}\frac{y}{x}+y^{-3}\tan^{-1}\frac{x}{y}$: first + second order?	$9u$.
81	[Class 005] $x^3\sin^{-1}\frac{y}{x}+x^4\tan^{-1}\frac{y}{x}$ at $(1,1)$?	$17\pi/2$.
82	[Class 005] Extensive / intensive degrees?	1 / 0.
83	[Class 006] Scaling test for degree n?	$u(\lambda x,\lambda y)=\lambda^n u(x,y)$.
84	[Class 006] Degree of $\frac{x^4+2x^2y^2+y^4}{(x^2+y^2)^2}$?	0 (it is identically 1).
85	[Class 006] Degree of $\frac{\sin^{-1}(y/x)}{\sqrt{x^2+y^2}}$?	-1.
86	[Class 006] Degree of $\frac{x^2y+xy^2}{(x+y)^3}$?	0.
87	[Class 006] Is $\frac{e^x-e^y}{x-y}$ homogeneous?	No.
88	[Class 006] Degree of $\frac{x+y}{\sqrt{x}+\sqrt{y}}$?	$\frac{1}{2}$.
89	[Class 006] $u=\frac{x-y}{x^2+y^2}$: xu_x+yu_y?	$-u$.
90	[Class 006] Same u: second order?	$2u=\frac{2(x-y)}{x^2+y^2}$.
91	[Class 006] Extension formula?	$xu_x+yu_y=nf(u)/f'(u)=g(u)$.
92	[Class 006] Extension, second order?	$g(u)[g'(u)-1]$.
93	[Class 006] $u=e^{x^3y^3/(x^2+y^2)}$: $f(u)$ and n?	$\log u$, $n=4$.
94	[Class 006] Same u: xu_x+yu_y?	$4u\log u$.
95	[Class 006] $u=\log\frac{x^2+y^2}{x+y}$: both sums?	1 and -1.
96	[Class 006] $u=\sin^{-1}\frac{x+y}{\sqrt{x}+\sqrt{y}}$: xu_x+yu_y?	$\frac{1}{2}\tan u$.
97	[Class 006] Same u: second order?	$\frac{1}{2}\tan u(\frac{1}{2}\sec^2 u-1)$.
98	[Class 006] $u=\sin^{-1}\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}$: $xu_x+yu_y+zu_z$?	$-3\tan u$.

99	[Class 006] $u = \tan^{-1}\frac{x^4+y^4}{xy}$: both sums?	$\sin 2u$; $\sin 2u(1 - 4\sin^2 u)$.
100	[Class 006] $u = x^3y^2\sin^{-1}(y/x)$: both sums?	$5u$; $20u$.
101	[Class 006] $x^3\sin^{-1}\frac{y}{x} + x^4\tan^{-1}\frac{y}{x}$, first + second at $(1,1)$?	$17\pi/2$.
102	[Class 006] $f(x/y, y/z, z/x)$: Euler sum?	0.
103	[Class 007] Taylor's series of $f(x)$ about $x = a$?	$f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \dots$
104	[Class 007] Maclaurin's series is Taylor's series about?	The origin.
105	[Class 007] Condition for Maclaurin's series?	f and all derivatives finite at 0 (series convergent).
106	[Class 007] e^x?	$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$
107	[Class 007] $\log(1+x)$, and where?	$x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$, $\lvert x\rvert < 1$.
108	[Class 007] $\sin x$?	$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$
109	[Class 007] $\cos x$?	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$
110	[Class 007] $\tan^{-1} x$?	$x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$
111	[Class 007] Does $1 + 2 + 4 + \dots$ equal -1?	No, it diverges; $\frac{1}{1-x}$ needs $\lvert x\rvert < 1$.
112	[Class 007] $\int_0^1 \sin(x^2)dx$?	$\frac{1}{3} - \frac{1}{42} + \frac{1}{1320} - \dots \approx 0.3103$.
113	[Class 007] $\lim \frac{\sin x - x}{x^3}$?	$-\frac{1}{6}$.
114	[Class 007] Two-variable second-order bracket?	$\frac{1}{2!}\{h^2 f_{xx} + 2hk f_{xy} + k^2 f_{yy}\}$.
115	[Class 007] Third-order bracket?	$\frac{1}{3!}\{h^3 f_{xxx} + 3h^2k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}\}$.
116	[Class 007] Why the 2 in $2hkf_{xy}$?	$f_{xy} = f_{yx}$.
117	[Class 007] First-degree Taylor polynomial is the ...?	Tangent plane.
118	[Class 007] $\sin x \sin y$ about $(\frac{\pi}{4}, \frac{\pi}{4})$, second order?	$\frac{1}{2} + \frac{1}{2}(h+k) - \frac{1}{4}(h-k)^2$.
119	[Class 007] Its third-degree term?	$-\frac{1}{12}(h+k)^3$.
120	[Class 007] $\log(1 + x - y)$ to third degree?	$(x-y) - \frac{(x-y)^2}{2} + \frac{(x-y)^3}{3}$.
121	[Class 007] $e^{ax}\sin by$ to third degree?	$by + abxy + \frac{a^2b}{2}x^2y - \frac{b^3}{6}y^3$.
122	[Class 007] $\sin(x + 2y)$ to third degree?	$(x+2y) - \frac{(x+2y)^3}{6}$.
123	[Class 007] $e^x\cos y$ about $(1, \frac{\pi}{4})$, second degree?	$\frac{e}{\sqrt{2}}[1 + h - k + \frac{1}{2}(h^2 - 2hk - k^2)]$.
124	[Class 007] $\tan^{-1}\frac{0.9}{1.1}$ by Taylor?	≈ 0.6854.
125	[Class 008] "In powers of $(x-1)$ and $(y - \frac{\pi}{4})$" means?	About $(1, \frac{\pi}{4})$.

126	[Class 008] $e^x \cos y$ derivatives at $(1, \frac{\pi}{4})$?	All $\pm\frac{e}{\sqrt{2}}$.
127	[Class 008] $e^x \cos y$ to second degree?	$\frac{e}{\sqrt{2}}[1 + h - k + \frac{1}{2}(h^2 - 2hk - k^2)]$.
128	[Class 008] $\log(2x + y + 1)$, three terms?	$2x + y - 2x^2 - 2xy - \frac{y^2}{2}$.
129	[Class 008] "Up to three terms" means?	Three non-zero terms.
130	[Class 008] $\tan^{-1}(xy)$ about $(1, -1)$?	$-\frac{\pi}{4} - \frac{1}{2}h + \frac{1}{2}k + \frac{1}{4}(h^2 + k^2)$.
131	[Class 008] $\tan^{-1}(0.9 \times (-1.2))$ to first order?	$-\frac{\pi}{4} - 0.05 = -0.835$.
132	[Class 008] ... to second order?	-0.823 (exact -0.824).
133	[Class 008] e^{x-y} about the origin?	$1 + (x - y) + \frac{(x-y)^2}{2} + \frac{(x-y)^3}{6}$.
134	[Class 008] $f_{xy}(0, 0)$ for $e^x \log(1 + y)$?	1 (the slide prints 0).
135	[Class 008] First six terms of $e^x \log(1 + y)$?	$y + xy - \frac{y^2}{2} + \frac{x^2 y}{2} - \frac{xy^2}{2} + \frac{y^3}{3}$.
136	[Class 008] $\sin(xy)$ about $(1, \frac{\pi}{2})$?	$1 - \frac{\pi^2}{8}h^2 - \frac{\pi}{2}hk - \frac{1}{2}k^2$.
137	[Class 008] $f_{yy}(1, 1)$ for $xy^2 + y\cos(x - y)$?	1 (the slide prints 3).
138	[Class 008] $xy^2 + y\cos(x - y)$ about $(1, 1)$?	$2 + h + 3k + \frac{1}{2}(-h^2 + 6hk + k^2)$.
139	[Class 008] $\sin x \sin y$ about $(\frac{\pi}{4}, \frac{\pi}{4})$, third degree?	$\frac{1}{2} + \frac{1}{2}(h + k) - \frac{1}{4}(h - k)^2 - \frac{1}{12}(h + k)^3$.
140	[Class 008] $\log(1 + x - y)$ to third degree?	$(x - y) - \frac{(x-y)^2}{2} + \frac{(x-y)^3}{3}$.
141	[Class 008] $e^{ax} \sin by$ to third degree?	$by + abxy + \frac{a^2 b}{2}x^2 y - \frac{b^3}{6}y^3$.
142	[Class 008] $\sin(x + 2y)$ to third degree?	$(x + 2y) - \frac{(x+2y)^3}{6}$.
143	[Class 008] $\tan^{-1}\frac{0.9}{1.1}$ by Taylor?	≈ 0.6854.
144	[Class 009] Relative maximum at (a, b)?	$f(a, b) > f(a + h, b + k)$ for all small h, k.
145	[Class 009] In terms of Δ?	$\Delta = f(a + h, b + k) - f(a, b)$ one sign: negative max, positive min.
146	[Class 009] Stationary points?	Solve $f_x = 0$, $f_y = 0$.
147	[Class 009] r, s, t?	f_{xx}, f_{xy}, f_{yy} at the point.
148	[Class 009] Maximum test?	$rt - s^2 > 0$ and $r < 0$.
149	[Class 009] Minimum test?	$rt - s^2 > 0$ and $r > 0$.
150	[Class 009] $rt - s^2 < 0$?	Saddle point.
151	[Class 009] $rt - s^2 = 0$?	Doubtful; needs further investigation.
152	[Class 009] What is the Hessian here?	$\begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix} = rt - s^2$.
153	[Class 009] Nature of $x^2 - y^2$ at the origin?	Saddle ($rt - s^2 = -4$).
154	[Class 009] $x^4 + y^4$ at the origin?	Test fails; minimum by Δ.

155	[Class 009] Min of $4x^2 + 2y^2 + 4xy - 10x - 2y - 3$?	$-\frac{23}{2}$ at $(2, -\frac{3}{2})$.
156	[Class 010] Stationary points of $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$?	$(4, 0)$, $(6, 0)$, $(5, \pm 1)$.
157	[Class 010] Its maximum and minimum?	Max 112 at $(4, 0)$; min 108 at $(6, 0)$.
158	[Class 010] Its $rt - s^2$?	$36[(x-5)^2 - y^2]$.
159	[Class 010] Stationary points of $x^3 - 12x + y^3 + 3y^2 - 9y$?	$(\pm 2, 1)$, $(\pm 2, -3)$.
160	[Class 010] Its extreme values?	Max 43 at $(-2, -3)$; min -21 at $(2, 1)$.
161	[Class 010] $x^4 + y^4 - x^2 - y^2 + 1$: max? min?	Max 1 at $(0, 0)$; min $\frac{1}{2}$ at $(\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}})$.
162	[Class 010] $\sin x + \sin y + \sin(x + y)$?	Max $\frac{3\sqrt{3}}{2}$ at $(\frac{\pi}{3}, \frac{\pi}{3})$.
163	[Class 010] Highest temperature $400xyz^2$ on the unit sphere?	50 at $x = y = \pm\frac{1}{2}$.
164	[Class 010] $kxyz^2$ on a sphere of radius a?	$\frac{ka^4}{8}$.
165	[Class 010] Open box of volume V, least area?	$x = y = \sqrt[3]{2V}$, height $\frac{x}{2}$.
166	[Class 010] Is a cube optimal for an open box?	No; only for a closed box.
167	[Class 011] Auxiliary function?	$F = f + \lambda\phi$.
168	[Class 011] Equations to solve?	$F_x = F_y = F_z = 0$ and $\phi = 0$.
169	[Class 011] Main advantage?	No need to eliminate a variable; extends to more variables and constraints.
170	[Class 011] Main disadvantage?	Does not tell maximum or minimum.
171	[Class 011] Largest box in a sphere?	A cube, half-edge $\frac{r}{\sqrt{3}}$, $V = \frac{8r^3}{3\sqrt{3}}$.
172	[Class 011] Largest box in an ellipsoid?	$V = \frac{8abc}{3\sqrt{3}}$.
173	[Class 011] Nearest points of $x^2 - z^2 = 1$ to the origin?	$(\pm 1, 0, 0)$.
174	[Class 011] Distances from the origin to $5x^2 + 6xy + 5y^2 = 8$?	Max 2, min 1.
175	[Class 011] Distances of $(3, 4, 12)$ from the unit sphere?	Max 14, min 12.
176	[Class 011] Open box, surface 432 cm²?	$12 \times 12 \times 6$ cm.
177	[Class 011] Divide 24 for max xy^2z^3?	4, 8, 12.
178	[Class 011] Highest $400xyz^2$ on the unit sphere?	50.
179	[Class 011] Extreme of $x^3 + 8y^3 + 64z^3$, $xyz = 1$?	24 at $(2, 1, \frac{1}{2})$.

Mnemonic (study aid). *"Subscripts walk forward, partials walk back."* z_{xy}: x then y; $\frac{\partial^2 z}{\partial y\,\partial x}$: read from the right, x then y.

Mnemonic (study aid). *"Fix the other, slope the trace."* To read f_x at a point, fix y, look at the trace, and take its slope.

Mnemonic (study aid). *"Multiply down, add across."* In a tree diagram, multiply the derivatives along each path from the top to the variable you want, then add the paths.

Mnemonic (study aid). *"Implicit: minus, top over the one you want."* $\dfrac{dy}{dx} = -\dfrac{f_x}{f_y}$: the variable you differentiate by (x) on top, the one you solve for (y) below.

Mnemonic (study aid). *"Order k brings the falling product."* First order n, second $n(n-1)$, third $n(n-1)(n-2)$; the coefficients follow the binomial row 1, 2, 1 or 1, 3, 3, 1.

Mnemonic (study aid). *"Undo, test, divide."* Undo the outer function to get $f(u)$, test $f(u)$ for its degree n, then divide: $g = n\,f/f'$.

Mnemonic (study aid). *"Binomial brackets."* Order n is $\frac{1}{n!}(h\partial_x + k\partial_y)^n f$: coefficients 1, 2, 1 at second order and 1, 3, 3, 1 at third.

Mnemonic (study aid). *"Series, table, values, substitute, check."* The five steps of every class 008 problem (Figure F33.1).

Mnemonic (study aid). *"Determinant first, then r."* Positive Hessian: r negative = hill (max), r positive = valley (min); negative Hessian = saddle; zero = doubt.

Mnemonic (study aid). *"Case, point, table, value."* Split the factored equation into cases, collect the points, tabulate r and $rt - s^2$, then substitute for the values.

Mnemonic (study aid). *"Add, differentiate, divide out λ, constrain."* $F = f + \lambda\phi$; $F_x = F_y = F_z = 0$; equate the expressions for λ; finish with $\phi = 0$.

4 Last 10 minutes before the exam

Top facts

1. $z = f(x, y)$: each $(x, y) \in D$ gives one real number; the graph is a surface.
2. Limit: the same value along every path; two paths with different values ⇒ no limit.
3. $\frac{2xy}{3x^2+y^2}$: 0 along $y = 0$, $\frac{1}{2}$ along $y = x$ ⇒ no limit at the origin.
4. Continuity: $f(a, b)$ defined, limit exists, $\lim f = f(a, b)$. Not continuous does not mean no limit.
5. $z_x = \lim_{h\to 0} \frac{f(x+h,y)-f(x,y)}{h}$: differentiate in x, everything else constant.
6. Notation $\frac{\partial z}{\partial x} = z_x = f_x$; ∂ by Legendre (1786), reintroduced by Jacobi (1841).
7. Clairaut: $f_{xy} = f_{yx}$ when the second partials are continuous.
8. $\frac{\partial^2 z}{\partial y \partial x} = z_{xy}$: subscripts left to right, ∂ right to left.
9. A function of two variables has 2 first-order, 4 second-order and 2^n nth-order partials; m variables give m^n.
10. The derivative of a single-valued function is the slope of the tangent to its curve.
11. $z = f(x, y)$ is a surface; points (x, y) on a curve in the xy-plane give a curve on the surface.
12. A surface has infinitely many tangents at a point.
13. $f_x(x_0, y_0)$ = slope of the tangent parallel to the x-axis (in the plane $y = y_0$).
14. $f_y(x_0, y_0)$ = slope of the tangent parallel to the y-axis (in the plane $x = x_0$).
15. Other directions: resolve into components; the resultant of $f_x\mathbf{i}$ and $f_y\mathbf{j}$ is the gradient.
16. $f_{xx} > 0$: concave up in x; $f_{yy} > 0$: concave up in y; negative means concave down.
17. f_{xy} = how the x-slope changes as you move in y.
18. Continuous mixed partials are equal: $f_{xy} = f_{yx}$.
19. Total derivative: $z = f(x, y)$, $x(t)$, $y(t)$ gives $dz/dt = z_x\, dx/dt + z_y\, dy/dt$ (ordinary d).
20. Chain rule: $x(u, v)$, $y(u, v)$ gives $z_u = z_x x_u + z_y y_u$, $z_v = z_x x_v + z_y y_v$ (partial ∂).
21. Draw the tree diagram and know the chain before writing any formula.
22. Total differential: $dz = f_x\, dx + f_y\, dy$; it approximates changes and nearby values.
23. Rates: decreasing means negative; constant means derivative 0; constraints give missing rates.
24. Implicit, two variables: $dy/dx = -f_x/f_y$ (slide 004·4 has it upside down).
25. Implicit, three variables: $z_x = -f_x/f_z$, $z_y = -f_y/f_z$, $y_x = -f_x/f_y$; check that the point is on the surface.
26. Unknown f: keep f_x, f_y as symbols; eliminate x, y from final answers in r, θ or s, t.
27. Exam favourites: $u_x + u_y + u_z = 0$ for $f(x - y, y - z, z - x)$; $z_x^2 + z_y^2$ identities under $u = x^2 - y^2$, $v = 2xy$ or $x = e^u \sin v$.
28. Slide errors to avoid: $dz/dt = 104t^{12}$ (not $60t^{12}$), $d\theta/dt = -0.0481$ (not 0.04831), $v_y = -3v_r + 3v_s$.
29. Homogeneous of degree n: $u = x^n g(y/x)$, equivalently $u(tx, ty) = t^n u$.
30. Degrees add for products and subtract for quotients; any degree is allowed.

31. Euler: $xu_x + yu_y = nu$ (three variables: add zu_z).
32. Corollary: $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$; with the first order added, n^2u.
33. Sums of parts with different degrees: apply each degree to its own part.
34. Extension: if $f(u)$ has degree n, $xu_x + yu_y = g = nf/f'$ and the second order is $g(g'-1)$.
35. $\tan^{-1}(\cdot)$ gives $g = \frac{n}{2}\sin 2u$; $\sin^{-1}(\cdot)$ gives $g = n\tan u$.
36. Degree 0 means a function of ratios only, so Euler's sum is 0; it does not mean constant.
37. ESA favourites: $5u$ & $20u$; $\sin 2u$ & $\sin 2u(1 - 4\sin^2 u)$; $9u$; $17\pi/2$.
38. Applications: scaling, economics, thermodynamics (extensive degree 1, intensive degree 0).
39. Scaling test: $u(\lambda x, \lambda y) = \lambda^n u$; degree may be positive, negative or zero.
40. No single λ^n factors out $\Rightarrow$ not homogeneous ($\frac{e^x - e^y}{x-y}$).
41. u homogeneous: nu, $n(n-1)u$, together n^2u; parts of different degrees separately.
42. $\frac{x-y}{x^2+y^2}$: $n = -1$, answers $-u$ and $+2u$.
43. $f(u)$ homogeneous: $g = nf/f'$; second order $g(g'-1)$.
44. $\sin^{-1} \to n\tan u$; $\tan^{-1} \to \frac{n}{2}\sin 2u$; $\log \to n$ (second order $-n$); $e^{(\cdot)} \to nu\log u$.
45. Slide 006·4: $-3\tan u$; slide 006·12: $4u\log u$; slide 006·14: $1, -1$; slide 006·16: $\frac{1}{2}\tan u$.
46. ESA favourites: $5u$ & $20u$; $\sin 2u$ & $\sin 2u(1 - 4\sin^2 u)$; $9u$; $17\pi/2$.
47. Taylor about a: $f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \ldots$; Maclaurin: about the origin.
48. Needs finite derivatives at the point; $\log(1+x)$, $(1+x)^n$, $\frac{1}{1-x}$ only for $|x| < 1$.
49. Two variables: $f + \{hf_x + kf_y\} + \frac{1}{2!}\{h^2f_{xx} + 2hkf_{xy} + k^2f_{yy}\} + \ldots$, $h = x - a$, $k = y - b$.
50. Third order: $\frac{1}{3!}\{h^3f_{xxx} + 3h^2kf_{xxy} + 3hk^2f_{xyy} + k^3f_{yyy}\}$ (asked in 4 of 6 ESA questions).
51. Method: derivative table, evaluate at the point, substitute order by order.
52. One combination $(x + 2y,\ x - y)$: use the one-variable series.
53. ESA: $\sin x \sin y$ about $(\frac{\pi}{4}, \frac{\pi}{4})$; $e^x \cos y$ about $(1, \frac{\pi}{4})$; $\log(1 + x - y)$; $e^{ax}\sin by$; $\sin(x + 2y)$.
54. First degree = tangent plane; more terms follow the surface further.
55. Method: general series → derivative table → values at the point → substitute → check.
56. $h = x - a$, $k = y - b$; second order $\frac{1}{2!}\{h^2f_{xx} + 2hkf_{xy} + k^2f_{yy}\}$; third order $\frac{1}{3!}\{h^3f_{xxx} + 3h^2kf_{xxy} + 3hk^2f_{xyy} + k^3f_{yyy}\}$.
57. $e^x \cos y$ at $(1, \frac{\pi}{4})$: $\frac{e}{\sqrt{2}}[1 + h - k + \frac{1}{2}(h^2 - 2hk - k^2)]$ (ESA May 2024).
58. Functions of one combination $(2x + y,\ x - y,\ x + 2y)$: use the one-variable series.
59. Slide errors: $f_{yy}(1,1) = 1$ (not 3) for $xy^2 + y\cos(x-y)$; $f_{xy}(0,0) = 1$ (not 0) for $e^x\log(1+y)$.
60. Approximation: expand about a nearby easy point; keep second order (-0.823 vs exact -0.824).
61. ESA: $\sin x \sin y$ about $(\frac{\pi}{4}, \frac{\pi}{4})$ (2nd and 3rd degree), $e^x\cos y$, $\log(1 + x - y)$, $e^{ax}\sin by$, $\sin(x + 2y)$.
62. Max at (a, b): $f(a,b) > f(a+h, b+k)$ nearby; min: $<$. Extreme value = max or min.
63. Stationary points: $f_x = 0$, $f_y = 0$ (horizontal tangent plane).
64. $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$; Hessian $rt - s^2$.

65. $rt - s^2 > 0$, $r < 0$: max. $rt - s^2 > 0$, $r > 0$: min.

66. $rt - s^2 < 0$: saddle point (not an extreme value). $rt - s^2 = 0$: doubtful.

67. Test every stationary point; substitute into f for the value.

68. Routine: $f_x = f_y = 0 \rightarrow$ points $\rightarrow r, s, t \rightarrow rt - s^2$ and $r \rightarrow$ values.

69. ESA $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$: max 112 at $(4, 0)$, min 108 at $(6, 0)$, saddles $(5, \pm 1)$.

70. ESA $x^3 - 12x + y^3 + 3y^2 - 9y$: max 43 at $(-2, -3)$, min -21 at $(2, 1)$, saddles $(2, -3)$, $(-2, 1)$.

71. $rt - s^2 < 0$: saddle, whatever the sign of r.

72. Constraint present: eliminate one variable (or use Lagrange, class 011).

73. $400xyz^2$ on the unit sphere: highest temperature 50.

74. Open box: base $\sqrt[3]{2V}$ square, height half the base edge.

75. $F = f + \lambda\phi$; solve $F_x = F_y = F_z = 0$, $\phi = 0$.

76. Make λ the subject (or multiply by x, y, z) to get relations like $x = y = z$.

77. Nature comes from the problem, not from the method.

78. Box in sphere ⇛ cube; in ellipsoid ⇛ $\frac{8abc}{3\sqrt{3}}$.

79. Distances: optimise the square. $5x^2 + 6xy + 5y^2 = 8$: 1 and 2.

80. $400xyz^2$ on the unit sphere: 50. Open box 432 cm²: $12 \times 12 \times 6$. 24 ⇛ 4, 8, 12.

Top formulas

E4.1 · Partial derivatives from the definition

ESA ★★

$$\frac{\partial z}{\partial x} = \lim_{h \to 0} \frac{f(x+h, y) - f(x, y)}{h}$$
$$\frac{\partial z}{\partial y} = \lim_{h \to 0} \frac{f(x, y+h) - f(x, y)}{h}$$

GIVES z_x (y held constant) and z_y (x held constant).

APPLIES When the limits exist.

NOTE Only one variable is moved by h; the other stays fixed.

S001·15

E4.2 · Notation for first-order partials

ESA ★★

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial x} = z_x = f_x, \qquad \frac{\partial z}{\partial y} = \frac{\partial f}{\partial y} = z_y = f_y$$

GIVES The equivalent symbols.

APPLIES $z = f(x, y)$.

NOTE The symbol is ∂ (“partial”, often read “dou”), not δ or d.

S001·15 · T001b 00:11:41 · T001b 00:13:05

E5.2 · Mixed partials

ESA ★★

$$z_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial y\,\partial x} = f_{xy}, \qquad z_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial x\,\partial y} = f_{yx}$$

GIVES The two crossed derivatives.

APPLIES Differentiate in one variable, then the other.

NOTE Subscripts: left to right. ∂ notation: right to left.

S001·16 · S001·17 · T001b 00:18:41

E5.3 · Clairaut's theorem

ESA ★★★

$$f_{xy} = f_{yx} \qquad \left(\frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial^2 f}{\partial x\,\partial y}\right)$$

GIVES The order of differentiation does not matter.

APPLIES f_{xy} and f_{yx} continuous near the point.

NOTE Called “a very important theorem” in the lecture; it is why Taylor's series has $2hkf_{xy}$.

S001·17 · T001b 00:19:57 · T001b 00:20:10

E2.2 · Two-path test (limit does not exist)

ESA ★

$$\lim_{\text{path 1}} f \neq \lim_{\text{path 2}} f \implies \lim_{(x,y)\to(a,b)} f(x,y) \text{ does not exist}$$

GIVES A proof that the limit does not exist.

APPLIES Substitution gives $\frac{0}{0}$.

NOTE Lines $y = mx$ through the origin are the usual first try: if the answer depends on m, the limit does not exist.

S001·10 · T001a 00:17:42

E3.1 · Continuity at (a, b)

ESA ★

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$$

GIVES The test for continuity.

APPLIES Three conditions: $f(a,b)$ defined, limit exists, both equal.

NOTE One variable: $\lim_{x\to c} f(x) = f(c)$.

S001·11 · T001b 00:00:38

E6.2 · Second-order partial derivatives ESA ★★

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \quad f_{yy} = \frac{\partial^2 f}{\partial y^2}, \quad f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y\,\partial x}, \quad f_{yx} = \frac{\partial^2 f}{\partial x\,\partial y}$$

GIVES The four second-order partials of $f(x, y)$.

APPLIES Differentiate twice. In f_{xy}, the subscripts are read left to right: first x, then y.

NOTE Notation order varies between books. The two mixed partials are equal when continuous (E6.4).

ADDED FOR CLARITY S002·3 · T002 00:00:58

E6.3 · How many partial derivatives of order n? ESA ★

2 variables: 2^n m variables: m^n

GIVES The number of nth-order partial derivatives, counting each order of differentiation separately.

APPLIES Two variables: 2 first-order, 4 second-order, 8 third-order.

NOTE With equal mixed partials, two variables give only $n + 1$ distinct ones.

S002·3 · T002 00:01:37 · T002 00:02:13

E9.1 · ∂f/∂x as an ordinary derivative ESA ★★

$$\left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)} = \left.\frac{d}{dx}\big(f(x, y_0)\big)\right|_{x=x_0}$$

GIVES Slope of the tangent to the surface at $(x_0, y_0, f(x_0, y_0))$ that is parallel to the x-axis.

APPLIES Fix $y = y_0$ first; $f(x, y_0)$ depends only on x.

NOTE The tangent lies in the plane $y = y_0$.

S002·7 · V002 00:07:45 · T002 00:06:10

E9.2 · ∂f/∂y as an ordinary derivative ESA ★★

$$\left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)} = \left.\frac{d}{dy}\big(f(x_0, y)\big)\right|_{y=y_0}$$

GIVES Slope of the tangent to the surface that is parallel to the y-axis.

APPLIES Fix $x = x_0$ first; $f(x_0, y)$ "just depends on y".

NOTE The tangent lies in the plane $x = x_0$.

S002·7 · V002 00:07:45 · T002 00:07:10

E10.1 · Gradient ESA ★

$$\nabla f = f_x\,\mathbf{i} + f_y\,\mathbf{j}$$

GIVES Vector whose components are the slopes along the x- and y-axes.

APPLIES Taken at the point (x_0, y_0).

NOTE Covered fully in vector calculus.

S002·8 · T002 00:09:23

E11.1 · Concavity from second partials ESA ★★

$$f_{xx} > 0 \Rightarrow \text{concave up in } x, \qquad f_{yy} > 0 \Rightarrow \text{concave up in } y$$

GIVES Shape of the surface along each axis direction.

APPLIES $f_{xx} < 0$ (or $f_{yy} < 0$) gives concave down in that direction.

NOTE Same test as d^2y/dx^2 for one variable.

S002·9 · T002 00:11:19

E14.1 · Total derivative rule ESA ★★

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

GIVES Rate of change of z with respect to the single variable t.

APPLIES $z = f(x, y)$, $x = x(t)$, $y = y(t)$.

NOTE Ordinary d on z, x, y with respect to t; partial ∂ only on z with respect to x, y.

S003·5 · T003a 00:05:54 · T003b 00:01:10

E15.1 · Total differential ESA ★★

$$dz = f_x(x, y)\,dx + f_y(x, y)\,dy$$

GIVES Approximate change in z for small changes dx, dy.

APPLIES $z = f(x, y)$ differentiable.

NOTE n variables: $dz = f_{x_1}dx_1 + f_{x_2}dx_2 + \cdots + f_{x_n}dx_n$.

S004·3 · S004·4 · S003·8 · T003a 00:13:56

E17.1 · Chain rule, two intermediate variables ESA ★★★

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}, \qquad \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$

GIVES Partial derivatives of a composite function.

APPLIES $z = f(x,y)$, $x = g(u,v)$, $y = h(u,v)$.

NOTE All derivatives are partial, because everything depends on two variables.

S003·13 · S003·14 · T003a 00:21:07

E18.1 · Implicit differentiation, two variables ESA ★★

$$f(x,y) = c \implies \frac{dy}{dx} = -\frac{f_x}{f_y}, \qquad f_y \neq 0$$

GIVES Slope of the level curve at a point.

APPLIES $f_y \neq 0$ at the point.

NOTE From $df = f_x\,dx + f_y\,dy = 0$. Slide 004·4 prints the fraction upside down.

S004·5 · S004·6 · T004a 00:06:36 · T003b 00:10:33

E19.1 · Implicit differentiation, three variables ESA ★★

$$f(x,y,z) = 0: \quad \frac{\partial z}{\partial x} = -\frac{f_x}{f_z}, \quad \frac{\partial z}{\partial y} = -\frac{f_y}{f_z}\ (f_z \neq 0); \qquad \frac{\partial y}{\partial x} = -\frac{f_x}{f_y}\ (f_y \neq 0)$$

GIVES Partial derivatives of the implicitly defined variable.

APPLIES Hold y constant for z_x; hold z constant for y_x.

NOTE Extends to any number of variables.

S004·6 · S003·26 · T004a 00:10:23 · T003b 00:09:29

E17.4 · Polar coordinates: derivatives you reuse ESA ★★

$$x = r\cos\theta,\ y = r\sin\theta: \qquad x_\theta = -r\sin\theta, \quad y_\theta = r\cos\theta$$
$$r = \sqrt{x^2+y^2},\ \theta = \tan^{-1}\tfrac{y}{x}: \quad r_x = \frac{x}{r},\ r_y = \frac{y}{r},\ \theta_x = -\frac{y}{r^2},\ \theta_y = \frac{x}{r^2}$$

GIVES The inner derivatives for polar problems.

APPLIES $r = \sqrt{x^2+y^2}$, $\theta = \tan^{-1}(y/x)$.

NOTE So $u_x = f_r\frac{x}{r} - f_\theta\frac{y}{r^2}$ and $u_y = f_r\frac{y}{r} + f_\theta\frac{x}{r^2}$ for $u = f(r,\theta)$.

S004·13 · S004·15 · S004·17

E20.1 · Homogeneous of degree n

ESA ★★

$$u = x^n g\left(\frac{y}{x}\right) = y^n h\left(\frac{x}{y}\right) \quad \Longleftrightarrow \quad u(tx, ty) = t^n\, u(x, y)$$

GIVES The definition and the equivalent scaling test.

APPLIES Two variables; with three, use $x^n g(y/x, z/x)$.

NOTE The scaling test is added for clarity; the slides use the $x^n g(y/x)$ form.

S005·3 · S005·4 · S005·5

E21.1 · Euler's theorem, two variables

ESA ★★★

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\,u$$

GIVES The scaled sum of first partials.

APPLIES u homogeneous of degree n.

NOTE Compact form: $xu_x + yu_y = nu$.

S005·6 · T005 00:08:57

E22.1 · Corollary: second order

ESA ★★★

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = n(n-1)\,u$$

GIVES The scaled sum of second partials.

APPLIES u homogeneous of degree n with continuous second partials.

NOTE Coefficients $1, 2, 1$ as in $(x+y)^2$.

S005·6 · T005 00:12:21

E22.3 · First plus second order

ESA ★★★

$$x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} + x\,u_x + y\,u_y = n(n-1)u + nu = n^2u$$

GIVES The combination asked in ESA Jan 2024 sets 3 and 4.

APPLIES u homogeneous of degree n; for $u = u_1 + u_2$ apply to each part.

NOTE Degrees 3 and -3 both give 9: so $9u$ for $x^3 \sin^{-1}\frac{y}{x} + y^{-3}\tan^{-1}\frac{x}{y}$.

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E23.1 · Extension, first order

ESA ★★★

$$f(u) \text{ homogeneous of degree } n \implies x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\frac{f(u)}{f'(u)} = g(u)$$

GIVES Euler's combination for a function that is not itself homogeneous.

APPLIES Some $f(u)$ is homogeneous (e.g. $\tan u$, $\sin u$, e^u).

NOTE $\tan u$: $g = n\tan u\cos^2 u = \frac{n}{2}\sin 2u$.

S005·7

E23.2 · Extension, second order

ESA ★★★

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = g(u)\left[g'(u) - 1\right]$$

GIVES The second-order combination for the extension.

APPLIES $g(u) = n\,f(u)/f'(u)$ from E23.1.

NOTE Differentiate g with respect to u.

S005·7